Retail Analytics Pipeline

Executive Summary

Project Purpose

The Retail Analytics Pipeline is a modular Python program that verifies the simulated transaction data from retail stores and calculates business metrics based on the performance of the branches. The program recognizes corrupt records before calculating the metrics for easy interpretation.

Design and Workflow

The program starts in **main.py**, which coordinates the pipeline. The validator checks required fields, parses ISO-format dates, converts item counts and revenue to numeric values, and flags missing or invalid data. Valid records are represented by the **TransactionRecord** class. The **RetailAnalytics** class calculates total sales from transaction objects. Transformation functions filter active transactions and calculate retention, top-performing branch, and sales by customer tier. The reporter module prints a formatted terminal summary. Automated unit tests are included.

Results from the Sample Run

Metric	Result
Raw records	3
Valid records	2
Active transactions	2
Total gross sales	195.50
Clean-record retention	66.67%
Top-performing branch	B001
Sales by customer tier	Gold: 120.50; Silver: 75.00

Interpretation

Two of the three sample records pass validation. The invalid record is flagged because its branch ID is missing. The two valid transactions total 195.50 in gross sales. Branch B001 has the higher branch total, and Gold customers account for 120.50 in sales.

Limitations

The program is designed using three hard-coded sample records instead of using an external file. The optional demonstration of net revenue utilizes tax rates of 13% and 15% for B001 and B002, respectively; however, tax rates have not been supplied by the assignment. The current data structure does not provide a region field and hence utilizes the branch ID as lookup key for these sample tax rates.